

TANGO: Temporal-Spatial Co-Scheduling for Multi-SLO LLM Serving

Abstract—Online large language model (LLM) serving processes each request through a prefill phase and a decode phase, whose latency is measured by time-to-first-token (TTFT) and time-per-output-token (TPOT), respectively. A key challenge is to meet heterogeneous Service-Level Objectives (SLOs) for both latency metrics while improving goodput, which is the rate of requests that meet both SLOs. However, existing LLM serving systems struggle to jointly adapt request priorities and the allocation of prefill and decode tokens within each execution step, leading to frequent SLO violations under heterogeneous latency requirements. We therefore propose TANGO, a temporal-spatial co-scheduling framework for Multi-SLO LLM serving. TANGO comprises a deadline-aware request dispatcher and a step-time-model-guided token allocator. The dispatcher prioritizes requests based on phase-aware SLO deadlines, while the token allocator regulates per-step workloads for prefill and decode phases using predicted execution time. Experimental results show that TANGO improves overall SLO attainment by 58.72%, and achieves 1.52× higher goodput on average than state-of-the-art systems.

I. INTRODUCTION

Online large language model (LLM) services consolidate requests from different applications on shared model instances to improve GPU utilization and reduce costs. These applications impose distinct latency requirements. For example, interactive chat prioritizes a fast first token generation, while real-time translation requires short latency between consecutive output tokens. These latency requirements correspond to two phases of LLM inference. During *prefill*, the model processes the prompt and generates the first output token, measured by the time to first token (TTFT). During autoregressive *decode*, it generates subsequent tokens one at a time, measured by the time per output token (TPOT).

Facing such a scenario, LLM serving should *improve the goodput while meeting the TTFT and TPOT Service-Level Objectives (SLOs)* for requests with heterogeneous latency requirements. To this end, current serving systems combine continuous batching, which dynamically admits new requests into the active batch between iterations, with chunked prefill, which divides long prompts into smaller segments that can be processed alongside decoding requests. Both techniques are adopted by state-of-the-art LLM serving engines such as vLLM and SGLang.

Fig. 1 shows the serving workflow for Multi-SLO requests under the prevailing LLM serving paradigm. For the requests in the waiting queue, at each execution step, *temporal scheduling* first determines which waiting requests to dispatch for execution alongside ongoing requests. Then, *spatial scheduling* allocates the step’s token budget among these requests to form a batch, specifying the sizes of prefill chunks and the number

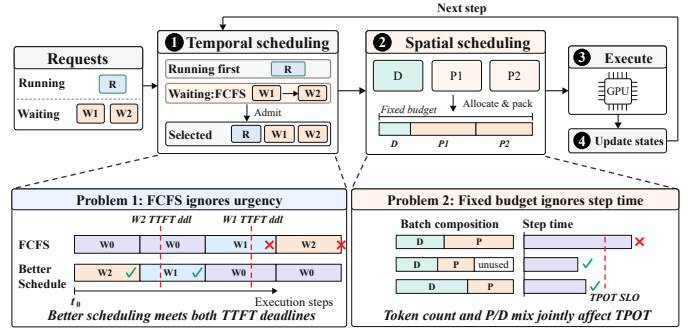


Fig. 1. Multi-SLO LLM serving from both temporal and spatial perspectives.

of decode tokens to include. At last, the engine executes the batch, updates request states, and repeats this process for the next step. The temporal and spatial scheduling jointly shape request queuing and step duration, affecting TTFT and TPOT.

Existing goodput-oriented serving engines commonly use first-come-first-served (FCFS) admission for temporal scheduling and a fixed maximum token budget for spatial scheduling, without appropriate adaptations to the heterogeneous SLOs of requests. From our investigations with vLLM on Qwen3-14B model using the mathematical-reasoning workload with heterogeneous TTFT and TPOT requirements, only 53.48% of requests satisfy both their TTFT and TPOT SLOs. The low SLO attainment stems from the inefficiencies from both the temporal and spatial scheduling aspect.

As for the temporal aspect, SLO-unaware request dispatching leads to an admission order that does not reflect request urgency. Under FCFS, a request with a tight TTFT objective can wait behind earlier requests with ample latency slack. The urgent request may exhaust its TTFT budget in the queue, even though the earlier requests could tolerate being served later. FCFS therefore misses the opportunity to defer less urgent requests and use their slack to reduce queueing delays for requests with tighter SLOs.

As for the spatial aspect, existing works allocate tokens under a fixed per-step budget without considering the TPOT requirements of active decode requests. However, we observe that the step time depends on both the total token count in a token batch and the prefill/decode composition. With 2,048 decode tokens fixed under Qwen3-14B model and RTX 6000 Ada GPU, adding 2,048 prefill tokens increases mean step time from 328.7 ms to 758.6 ms. Even with the total token count fixed at 4,096, pure-prefill, pure-decode, and evenly mixed batches take 610.5 ms, 682.5 ms, and 758.6 ms, respectively.

A fixed token budget alone therefore cannot reliably control step time to meet the TPOT requirements of active decode requests.

To adress these two inefficiencies, an appropriate solution is to coordinate request-level urgency at the temporal perspective, with batch-level execution time at the spatial perspective. However, two challenges still exist to achieve this solution.

First, temporal scheduling should coordinate request dispatching and preemption, deciding which waiting requests to admit and which ongoing requests to defer under resource contention. These requests have different active SLOs and remaining latency budgets, thus ordering them requires a consistent comparison of urgency in the prefill and decode phases. Second, spatial scheduling should balance prefill and decode latency requirements through per-step token allocation in a mixed token batch. More prefill token allocation advances urgent TTFT requests but delay decode requests, while fewer prefill tokens have the opposite effect. This tradeoff requires accurate step-time estimates for different prefill/decode compositions and context lengths to guide the token allocation.

We therefore propose TANGO, a multi-SLO LLM serving system that jointly controls temporal request dispatching and spatial token allocation. TANGO aims to improve the goodput while satisfying heterogeneous TTFT and TPOT SLO requirements of requests. It comprises a deadline-aware request dispatcher and a step-time-model-guided token allocator. The deadline-aware request dispatcher assigns phase-aware latency deadlines according to each request’s active SLO and uses them to guide admission and preemption, prioritizing requests with tighter latency budgets. The step-time-model-guided token allocator uses a step-time model to estimate the execution time of mixed prefill/decode batches, and accordingly adjusts their prefill token budgets according to the remaining latency slack of decoding requests.

This paper makes three major contributions:

- **Investigating the scheduling inefficiencies of current multi-SLO LLM serving.** The analysis motivates us to account for heterogeneous request urgency and mixed prefill/decode interference through joint control of temporal request dispatching and spatila token allocation.
- **The design of a deadline-aware request dispatcher that prioritizes requests across LLM prefill and decode phases.** By mapping each request’s active SLO to a comparable deadline, it guides request admission and preemption, mitigating temporal inefficiency caused by SLO-unaware queueing.
- **The design of a step-time-model-guided token allocator that adapts token batch composition to latency constraints.** Based on predicted step time, it converts the remaining TPOT slack into a prefill token budget, mitigating spatial mixed prefill/decode interference while preserving progress for requests with tight TTFT SLOs.

We evaluate TANGO against state-of-the-art LLM serving systems, including vLLM, SOLA, and JITServe, using Llama-3.1-8B, Qwen3-14B, Qwen3-32B, Llama-3.1-70B, and

TABLE I
SCHEDULING CAPABILITIES OF REPRESENTATIVE COLOCATED SERVING SYSTEMS.

System	Continuous Batching	Chunked Prefill	Heterogeneous SLOs	Request-Specific SLO Prioritization	Deadline-Aware Token Allocation
vLLM	✓	✓	–	–	–
Sarathi-Serve	✓	✓	–	–	–
SOLA	✓	✓	–	–	–
JITServe	✓	✓	✓	✓	–
TANGO	✓	✓	✓	✓	✓

Qwen3-Next-80B-A3B-Instruct under chatbot, mathematical-reasoning, and software-agent workloads. Compared to the baselines, TANGO improves the overall SLO attainment by 58.72% and improve the goodput by $1.52\times$ on average.

II. RELATED WORK

SLO-Agnostic LLM Serving. Existing LLM serving systems improved throughput and latency through efficient batching and memory management. Orca introduced iteration-level scheduling, which allowed requests to enter and leave a batch between decoding iterations. vLLM introduced PagedAttention to reduce KV-cache fragmentation and support flexible memory sharing. SGLang exploited prefix reuse through RadixAttention and optimized the execution of structured language model programs. However, these works optimized batching and memory efficiency without explicitly accounting for each request’s TTFT and TPOT objectives in scheduling.

SLO-Aware Temporal Request Scheduling. SLO-aware temporal schedulers used latency requirements to decide which requests received service. SOLA adjusted request priorities and iteration workloads based on request progress and system state to improve SLO attainment for both TTFT and TPOT. However, its shared TTFT and average-TPOT targets could not directly express heterogeneous request SLOs, and its token budgeting did not explicitly enforce individual next-token deadlines. For requests with heterogeneous SLOs, JITServe progressively refined estimates of output lengths and request dependencies to prioritize requests by their expected goodput and the serving bandwidth required to meet their SLOs, and grouped requests with similar input lengths to improve batching efficiency. However, this request-specific SLO prioritization operated at the inter-request level without explicitly accounting for intra-batch prefill–decode contention.

Spatial Prefill–Decode Scheduling. Existing approaches mitigated prefill–decode interference through resource separation or control of batch workloads. DistServe placed prefill and decode on separate GPUs to eliminate interference and independently optimized their resource allocation and parallelism to meet TTFT and TPOT requirements. Sarathi-Serve instead split long prompts into smaller prefill chunks and batched them with ongoing decode requests under a fixed token budget to reduce interference. However, these approaches relied on phase-level resource partitioning or fixed token budgets, limiting their ability to accommodate heterogeneous TTFT and TPOT requirements.

Table I compares TANGO with representative works.

III. BACKGROUND AND MOTIVATION

In this section, we first introduce the background of Multi-SLO serving and the traces for investigation. Then, we present the low Multi-SLO serving performance of existing systems and analyze the underlying reasons.

A. Multi-SLO LLM Serving

LLM Inference and Latency SLOs. LLM inference comprises two phases: *prefill* processes prompt tokens in parallel, builds key-value (KV) caches, and produces the first output token; *decode* reuses these caches to generate subsequent tokens autoregressively. The two phases expose different forms of latency and corresponding service-level objectives (SLOs): a request should first wait for its prompt to be processed, and should then make timely progress over a sequence of decoding iterations. Let request r_i arrive at time a_i and generate n_i output tokens, where $t_{i,j}$ denotes the time at which its j -th output token is returned. Its time to first token (TTFT) and time per output token (TPOT) are defined as

$$\begin{aligned} \text{TTFT}_i &= t_{i,1} - a_i, \\ \text{TPOT}_i &= \frac{1}{n_i - 1} \sum_{j=2}^{n_i} (t_{i,j} - t_{i,j-1}), \quad n_i > 1. \end{aligned} \quad (1)$$

TTFT captures the queuing and prefill delay before a response begins, whereas TPOT captures the average interval between consecutive tokens after generation starts. Each request is associated with a pair of latency objectives, denoted by S_i^{TTFT} and S_i^{TPOT} . Request r_i satisfies its SLOs only if $\text{TTFT}_i \leq S_i^{\text{TTFT}}$ and $\text{TPOT}_i \leq S_i^{\text{TPOT}}$. In this work, *Multi-SLO* refers both to these phase-specific objectives and to their heterogeneous values across requests and applications. Accordingly, serving efficiency is measured by *goodput*, i.e., the rate of requests that satisfy both objectives.

Continuous Batching and Chunked Prefill. Unlike static batching, continuous batching updates the active batch at iteration boundaries. After each iteration, completed requests leave, unfinished decode requests remain active, and waiting requests can be admitted into the released capacity. The batch composition can thus evolve throughout generation, improving GPU utilization while allowing the scheduler to reconsider which requests should run at every step.

A prefill request may contain hundreds or thousands of prompt tokens, and scheduling an entire prompt at once can therefore produce a long iteration, thus delay token generation for the decode requests in the same batch. Chunked prefill divides the prompt into smaller chunks that may be processed over multiple iterations. This enables prefill chunks and decode tokens to be colocated in one batch, while exposing the prefill chunk size as a controllable scheduling variable. Specifically, let $\mathcal{P}_k$ and $\mathcal{D}_k$ denote the prefill and decode requests selected in step k , and $c_{i,k}$ the prompt tokens allocated to request r_i . The step's total token count is

$$T_k = \sum_{r_i \in \mathcal{P}_k} c_{i,k} + |\mathcal{D}_k|, \quad T_k \leq B_{\max}, \quad (2)$$

where each decode request contributes one token and $B_{\max}$ is the maximum per-step token budget. The resulting *queuing delay* and *per-step interval between decoding iterations* decide whether each request meets its TTFT and TPOT SLOs.

B. Multi-SLO Traces for Investigations

We construct three representative LLM inference workloads.

Chatbots. We construct chatbot requests from LMSYS-Chat-1M, which contains one million real-world conversations collected from interactions with 25 LLMs. The token length of each recorded response determines its target output length.

Mathematical Reasoning. We use all 8,792 questions from the training and test splits of GSM8K. Each word problem serves as the input prompt, while the token length of its reference solution determines the target output length.

Software Agents. We use the software-engineering subset of AgentWorldBench and retain one request from each of 99 distinct trajectories. For each request, we generate 1,024 variants by prepending a distinct synthetic prefix while preserving the original agent action and target environment observation.

From the three request pools, we randomly sample 10,000 chatbot, 8,000 mathematical-reasoning, and 4,000 software-agent requests to form the replay traces.

Request arrivals and SLOs. We associate requests in order with timestamps from failure-free entries in BurstGPT and replay each workload according to these timestamps, thereby preserving the trace's temporal arrival pattern. For each workload, we use λ_{90} , the sending rate at which vLLM achieves 90% SLO attainment, as a baseline and replay at $1.2\lambda_{90}$ to represent a workload under pressure.

For each request, we execute it alone on an otherwise idle vLLM deployment and record its measured TTFT and TPOT. These isolated measurements serve as request-specific latency baselines. We derive the two SLOs of request r_i as

$$S_i^{\text{TTFT}} = c_i^{\text{TTFT}} \widehat{\text{TTFT}}_i, \quad S_i^{\text{TPOT}} = c_i^{\text{TPOT}} \widehat{\text{TPOT}}_i, \quad (3)$$

where $\widehat{\text{TTFT}}_i$ and $\widehat{\text{TPOT}}_i$ are the isolated-run measurements, and c_i^{TTFT} and c_i^{TPOT} are the coefficients assigned to the request's SLO class. Using a fixed random seed, we assign requests equally among four SLO classes: TTFT-prioritized, TPOT-prioritized, balanced, and relaxed. For Qwen3-14B, the TTFT/TPOT coefficient pairs for these four classes are (20, 2.5), (50, 1.5), (25, 2), and (50, 3), respectively.

All investigation systems use the same request sequence, arrival schedule, sending rate, and SLO targets under each experimental condition. Other sending rates and SLO settings are also explored in Sections VIII-F and VIII-G.

C. Low Multi-SLO Serving Performance

We use vLLM's default FCFS scheduler as a representative serving baseline for investigations. It supports continuous batching and chunked prefill, but schedules requests without differentiating their TTFT and TPOT SLOs. We run Qwen3-14B with $\text{TP} = 2$ on two GPUs using the configuration in Table II. We use the math workload as a representative case and replay it at its stress load of $1.2\lambda_{90}$.

TABLE II
EXPERIMENTAL SETUP.

	Specifications
Hardware	2× Intel Xeon Gold 6530 CPUs, 64 physical cores 128 logical threads, 256 GiB host RAM 4× RTX 6000 Ada GPUs (48 GB each)
Software	meta-llama/Llama-3.1-8B, Qwen/Qwen3-14B, Qwen/Qwen3-32B ; LMSYS-Chat-1M, GSM8K, AgentWorldBench-SWE, BurstGPT

TABLE III
SLO ATTAINMENT OF VLLM ON THE MATH WORKLOAD.

SLO Class	TTFT (%)	TPOT (%)	Joint (%)
Overall	75.80	77.68	53.48
TTFT-prioritized	48.89	100.00	48.89
TPOT-prioritized	100.00	12.00	12.00
Balanced	57.39	100.00	57.39
Relaxed	100.00	100.00	100.00

Table III shows the SLO attainment rates for the math workload. Under FCFS, the overall TTFT and TPOT attainment rates are 75.80% and 77.68%, respectively, but only 53.48% of them satisfy both SLO objectives. The gap between individual and joint attainment shows that satisfying either latency SLO alone does not provide effective Multi-SLO service.

The class-level results further reveal distinct failure modes. TTFT-prioritized requests achieve 48.89% TTFT attainment and 100.00% TPOT attainment, whereas TPOT-prioritized requests achieve 100.00% and 12.00%, respectively. Balanced and relaxed requests attain both objectives for 57.39% and 100.00% of their requests, respectively. The contrasting class-level results indicate that the low joint attainment cannot be attributed to a single global latency bottleneck.

The above results reveal a mismatch between throughput-oriented scheduling and request-specific latency SLOs. FCFS does not account for the remaining TTFT or TPOT slack when admitting prefill chunks or advancing decode requests. A scheduling decision that preserves prefill prompt responsiveness may disrupt decode token delivery, and vice versa. The serving engine can thus retain sufficient capacity to satisfy 100.00% of the relaxed requests, yet fail to direct that capacity toward the phase-specific objectives of other requests. These observations motivate a closer examination of how request ordering in the temporal aspect and per-step token allocation in the spatial aspect jointly affect Multi-SLO attainment.

D. Diving into Underlying Reasons

Our investigation identifies the following two complementary causes of the low Multi-SLO attainment from the perspective of temporal and spatial scheduling, respectively.

1) *Temporal Request Dispatch with TTFT SLO Unawareness*: The first-token latency of request r_i is decomposed as

$$\text{TTFT}_i = W_i^{\text{queue}} + L_i^{\text{prefill}}, \quad (4)$$

where W_i^{queue} is the delay from arrival until the request is first admitted for prefill, and L_i^{prefill} is the remaining latency until its first token is produced. For a chunked prefill, L_i^{prefill}

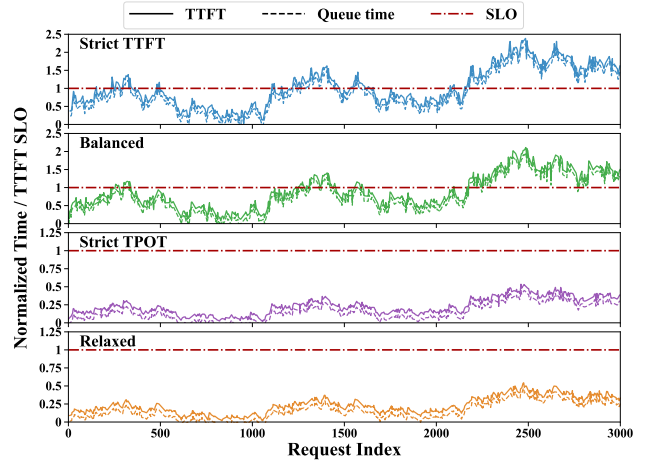


Fig. 2. Normalized TTFT and queueing time under FCFS. **TTFT-prioritized**

includes the execution and intervening scheduling delays of all prompt chunks after admission.

Fig. 2 plots TTFT and queueing time for the first 3,001 math requests, normalized by each request’s TTFT SLO. Solid and dashed lines denote TTFT and queueing time, respectively, and the dash-dotted line marks the SLO boundary. We can observe that the normalized queueing-time curve closely follows the TTFT curve in all four SLO classes, indicating that queueing latency dominates. For TTFT-prioritized and balanced requests, the queueing latency frequently consumes most, or even all, of the TTFT budget before prefill completes. TPOT-prioritized and relaxed requests remain below the boundary because their looser TTFT objectives provide substantially more slack. The resulting TTFT violations therefore arise primarily from how waiting time is distributed among requests, rather than from a uniform shortage of prefill compute capacity.

The TTFT-SLO-unaware dispatching (e.g., FCFS) creates this imbalance because it admits requests according to arrival order, regardless of their remaining TTFT slack. A TTFT-prioritized request may wait behind earlier requests whose looser objectives could tolerate additional delay.

The temporal scheduling problem is thus to determine *which request* should receive prefill service first. Accounting for the remaining TTFT slack can redistribute queueing latency toward requests that can tolerate it, without changing the total amount of prefill work.

2) *Spatial Token Allocation with a Fixed Budget*: For an active decode request, the time interval before its next token is determined by the duration of the engine step in which that token is scheduled. Reordering waiting requests alone cannot protect TPOT because a decode token may still share a step with a large prefill chunk.

To quantify this interference, we vary the numbers of prefill tokens p and decode tokens d scheduled in the same step and measure the mean step time, as shown in Fig. 3. Each case is measured on Qwen3-14B with TP = 2 over 50 timed steps (5 rounds × 10 measurements), and gray cells exceed the maximum per-step token budget. Shown in the blue outline,

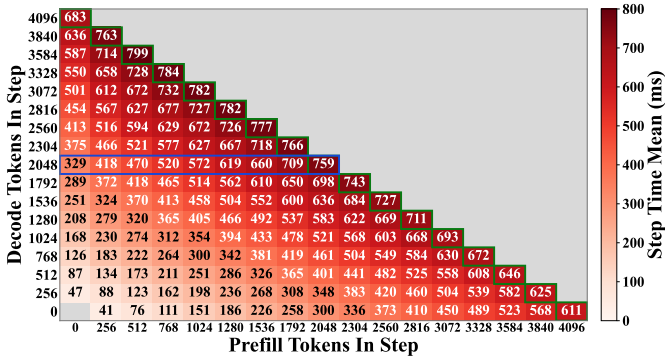


Fig. 3. Mean step time with different prefill and decode token counts.

when $d = 2,048$, increasing p from 0 to 2,048 increases the mean step time from 329 ms to 759 ms. Every active decode request has to wait for the longer step to complete before receiving its next token, directly increasing its TPOT.

More importantly, the total token count alone does not determine step time. Along the green boundary where $p + d = 4,096$, the pure-prefill and pure-decode configurations take 611 ms and 683 ms, respectively, while mixed configurations reach up to 799 ms. Thus, the same total token budget produces about a $1.3\times$ difference in latency. This nonlinearity reflects the distinct execution characteristics of compute-intensive prefill and memory-intensive decode, together with their interference when colocated in one iteration. Consequently, enforcing only $p + d \leq B_{\max}$ bounds the amount of scheduled work but cannot reliably bound the wall-clock latency experienced by active decode requests.

The spatial scheduling problem is thus to determine *how many prefill tokens* can safely share the next step with the current decode workload. This requires an estimate of mixed-step latency and a mechanism that converts the remaining TPOT slack of active requests into a safe prefill allocation.

Temporal request ordering and spatial token allocation address different failure modes: optimizing either dimension alone leaves the other source of SLO violations unresolved. These observations motivate their joint design in TANGO.

IV. TANGO METHODOLOGY

Fig. 4 shows the design overview of TANGO, which comprises a deadline-aware request dispatcher and a step-time-model-guided token allocator. At each iteration boundary, the two components coordinate request execution priority and per-step token allocation according to heterogeneous TTFT and TPOT SLOs. They take each request’s SLOs, inference phase, and execution progress as input and produce a mixed batch of prefill chunks and decode tokens.

The temporal scheduling challenge is to compare request urgency across inference phases. A request with an unfinished prefill must produce its first token within its TTFT budget, whereas a decoding request must sustain token-generation progress within its TPOT budget. The dispatcher maps these

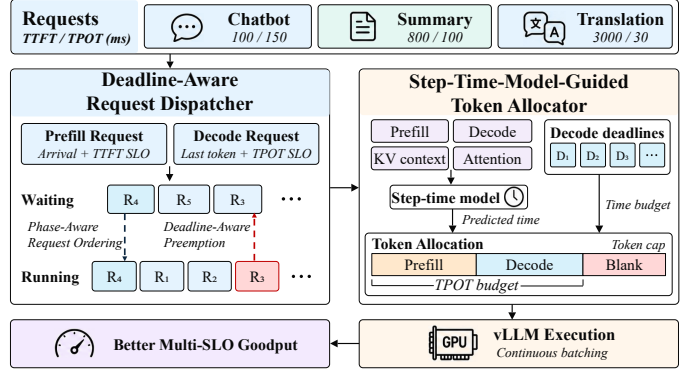


Fig. 4. Design overview of TANGO. ~~del~~ CHN

phase-specific requirements to comparable deadlines and updates request priorities as inference progresses. These priorities guide the admission of waiting requests and the preemption of less urgent ongoing requests under resource contention (Section V).

The spatial scheduling challenge is to advance urgent prefills while limiting their interference with decoding requests. As shown in Section III-D, the total token count alone cannot reliably determine mixed-step execution time. The allocator therefore uses a step-time model to predict the duration of candidate batches from their prefill/decode composition and context-dependent attention work. It converts the remaining deadline slack of selected decode requests into a prefill token budget and distributes that budget according to the dispatcher’s priorities (Section VI).

Specifically, TANGO operates as follows. First, the scheduler collects the running requests’ execution state, and the allocator derives a prefill token budget using the step-time model. Second, the dispatcher orders waiting requests by their phase-aware deadlines for admission alongside ongoing requests. If a waiting request cannot obtain KV-cache blocks, the dispatcher considers preempting the running request with the latest deadline, but proceeds only when the waiting request has an earlier deadline. The allocator applies the remaining token and time budgets to the prefill chunks of ongoing and newly admitted requests. Finally, the engine executes the resulting batch and returns token timestamps and request progress to update the next scheduling decision. Unfinished requests remain eligible for reconsideration at the next iteration boundary.

The example in Fig. 4 illustrates how the two components cooperate. An urgent waiting request r_4 receives earlier admission, while a less urgent ongoing request can be preempted to release capacity. The allocator then determines how much of the admitted request’s prompt can share a step with the active decode workload. When the decode requests have little remaining deadline slack, it limits the prefill chunk size and leaves the remaining prompt tokens for subsequent steps, even if the engine’s maximum token capacity has not been exhausted. Thus, request urgency determines the service order, while predicted step time determines how much work can execute in that order.

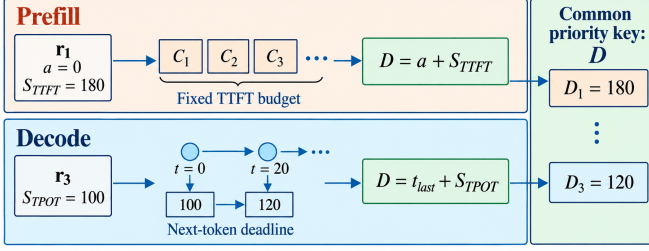


Fig. 5. Phase-aware deadlines for request prioritization (times in ms).

V. DEADLINE-AWARE REQUEST DISPATCHER

Figs. 5 and 6 illustrate the dispatcher, which uses phase-aware deadlines as a common priority signal for request admission and preemption. In this section, we first introduce phase-aware deadline construction (Fig. 5), then explain how these deadlines guide waiting-queue ordering and preemption under KV-cache memory pressure (Fig. 6).

A. Phase-Aware Request Ordering

As analyzed in Section III-D, arrival order does not reflect the different latency budgets of waiting requests. Moreover, a request's active latency objective changes during inference: before its first output token, it must produce that token within its TTFT budget; afterward, it must make timely progress in decoding. Our key idea is to express these phase-specific objectives as an effective deadline on the same time axis. Let q_i denote the number of output tokens already returned by request r_i . Using the timestamps and SLOs defined in Section III-A, its current deadline is

$$D_i = \begin{cases} a_i + S_i^{\text{TTFT}}, & q_i = 0, \\ t_{i,q_i} + S_i^{\text{TPOT}}, & q_i > 0. \end{cases} \quad (5)$$

The first case bounds the time available before the first token, while the second gives a rolling target for the next token based on the most recent output timestamp, denoted by t_{last} in Fig. 5. Both cases map the active latency objective to the common priority key D , allowing the dispatcher to compare requests across inference phases.

The deadline is maintained according to output progress rather than reset whenever a request is scheduled. During chunked prefill, all chunks share the original TTFT deadline, so both queueing and the execution of successive chunks consume the same latency budget. For example, r_1 in Fig. 5 arrives at 0 ms with a TTFT SLO of 180 ms; its deadline remains 180 ms across all prompt chunks. When the first token is returned, the active objective switches to TPOT. Each subsequent output token updates the timestamp in Eq. 5 and establishes a new next-token deadline. For r_3 with a TPOT SLO of 100 ms, advancing the latest output timestamp from 0 to 20 ms moves its deadline from 100 to 120 ms. This rolling deadline provides a local target for each inter-token interval; meeting these targets is sufficient to satisfy the average-TPOT objective in Eq. 1.

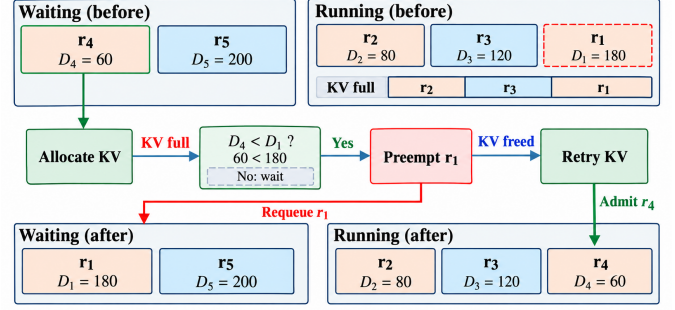


Fig. 6. Deadline-aware admission and preemption (times in ms).

At each iteration boundary, the dispatcher orders waiting requests by increasing D_i and admits them in earliest-deadline-first (EDF) order. Arrival time breaks deadline ties, with request identifiers providing deterministic ordering when both values are equal. The waiting queue contains both newly arrived requests and preempted requests awaiting resumption. A request that has not returned its first token is ranked by its TTFT deadline, while a resumed decode request is ranked by its TPOT deadline, even if recovering its context requires additional prefill work. Admission remains subject to the engine's sequence and memory limits and the token budget supplied by the allocator. In Fig. 6, r_4 with $D_4 = 60$ ms is considered before r_5 with $D_5 = 200$ ms. This ordering allows requests with later deadlines to absorb more queueing delay, giving earlier service to more urgent requests without changing their total prefill work.

B. Deadline-Aware Preemption

During continuous batching, a new or resumed request may fail to obtain the KV-cache blocks required for admission. The dispatcher then uses the same deadline representation to identify a running request that can yield memory. For the current running set $\mathcal{R}_k$, it selects the candidate victim as

$$v = \arg \max_{i: r_i \in \mathcal{R}_k} (D_i, a_i), \quad (6)$$

where the tuple is compared lexicographically, so a later-arriving request is selected when deadlines are equal. For waiting request r_j , preemption proceeds only if $D_j < D_v$. Otherwise, r_j remains queued, preserving the running requests whose deadlines are equally or more urgent.

When the comparison permits preemption, the engine reclaims the victim's KV-cache blocks, returns it to the waiting queue, and retries admission of r_j . Preemption occurs at an iteration boundary and preserves the victim's SLOs, arrival time, output-token count, and latest output timestamp. The victim re-enters the queue with the deadline of its current phase: its original TTFT deadline before the first token, or its existing next-token deadline afterward. Thus, preemption does not reset the latency budget, and waiting for resumption continues to consume its remaining slack. In the example at $t = 40$ ms in Fig. 6, KV allocation for r_4 fails, and r_1 has the

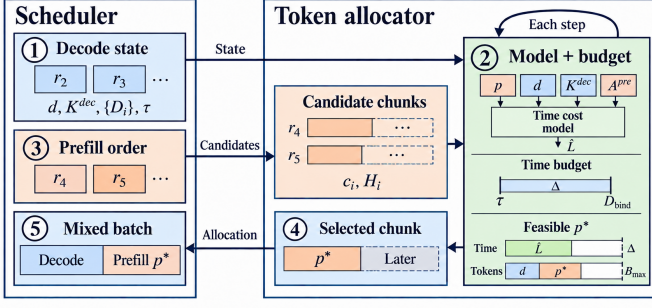


Fig. 7. Execution workflow of the step-time-model-guided token allocator.

latest deadline among the running requests. Since $D_4 = 60$ ms is earlier than $D_1 = 180$ ms, the dispatcher preempts r_1 and admits r_4 after retrying allocation. Request r_1 retains its 180 ms deadline and returns to the waiting queue ahead of r_5 , while r_2 and r_3 remain running.

VI. STEP-TIME-MODEL-GUIDED TOKEN ALLOCATOR

Fig. 7 shows how the allocator determines the amount of prefill work that can share an engine step with the active decode workload. Its key idea is to translate the remaining next-token time budget into a prefill token allocation using a fitted step-time model. In this section, we first introduce the model, then explain how heterogeneous TTFT and TPOT objectives guide budgeting, and finally describe the allocation workflow.

A. Step-Time Modeling

As analyzed in Section III-D, batches with the same total token count can have substantially different execution times. The predictor must therefore account for both batch composition and context-dependent attention work. Let p and d denote the numbers of prefill and decode tokens, respectively, K^{dec} the sum of context lengths accessed by decode requests, and A^{pre} the effective attention work of prefill chunks. These four inputs feed the time cost model in Fig. 7, which estimates the step duration as

$$\hat{L}(p, d, K^{\text{dec}}, A^{\text{pre}}) = \alpha + \beta p + \gamma d + \delta K^{\text{dec}} + \epsilon A^{\text{pre}}. \quad (7)$$

Each term approximates a different source of execution cost. The intercept α captures the fixed runtime cost of a step, while βp and γd capture the token-dependent costs of prefill and decode computation. The term δK^{dec} accounts for decode attention's KV-cache reads, which grow with the total history accessed by decode tokens. Finally, ϵA^{pre} accounts for the attention computation performed by prefill chunks.

For chunked prefill, the attention work depends on both the new chunk and its existing prefix. Let $H_{i,k}$ be the already-computed context length available to request r_i before step k . Using the chunk sizes $c_{i,k}$ defined in Eq. 2, we approximate the attention work as

$$A_k^{\text{pre}} = \sum_{r_i \in \mathcal{P}_k} c_{i,k} \left(H_{i,k} + \frac{c_{i,k}}{2} \right). \quad (8)$$

A single fresh prefill thus contributes approximately $p^2/2$, whereas a chunk following a long prefix incurs additional work proportional to that prefix length. Consequently, equal-sized chunks can consume different amounts of the step-time budget. The coefficients are fitted from profiled engine steps for the deployed model and hardware configuration. At runtime, prediction requires only these four workload statistics and a few arithmetic operations.

B. Joint TTFT/TPOT Budgeting

At the beginning of step k , the allocator summarizes the active decode workload as $d_k = |\mathcal{D}_k|$ and $K_k^{\text{dec}} = \sum_{r_i \in \mathcal{D}_k} H_{i,k}$. Each decode request contributes its next-token deadline D_i from Eq. 5. The allocator selects a binding deadline from this set using a configurable percentile $\rho_0 \in [0, 1]$. With $\rho_0 = 0$, the earliest deadline bounds the predicted step duration; a larger percentile selects a later deadline and permits more prefill work.

A fixed decode percentile can leave urgent prefills with insufficient service even when increasing prefill work would improve first-token responsiveness. To account for this trade-off, the allocator compares the remaining fractions of the two latency budgets. At scheduling time τ_k , these fractions are

$$u_{i,k}^{\text{TTFT}} = \frac{D_i - \tau_k}{S_i^{\text{TTFT}}}, \quad q_i = 0, \\ u_{i,k}^{\text{TPOT}} = \frac{D_i - \tau_k}{S_i^{\text{TPOT}}}, \quad q_i > 0. \quad (9)$$

The allocator compares the minimum TTFT fraction among requests awaiting their first token in the running set and waiting queue with the minimum TPOT fraction among active decode requests. If TTFT is more urgent, it moves the active percentile ρ_k from ρ_0 toward 1, allowing a later decode deadline to release additional prefill capacity. Otherwise, it retains $\rho_k = \rho_0$. This adaptation gives urgent prefills more service by relaxing the step-time target imposed by earlier decode deadlines.

Let Quantile_{ρ_k} select a deadline at percentile ρ_k from the ordered decode deadlines. Denote the selected deadline by D_k^{bind} , shown as D_{bind} in Fig. 7. The time available for the next step is

$$D_k^{\text{bind}} = \text{Quantile}_{\rho_k}(\{D_i : r_i \in \mathcal{D}_k\}), \\ \Delta_k = D_k^{\text{bind}} - \tau_k. \quad (10)$$

The full prediction in Eq. 7 is compared with Δ_k , since fixed and decode-related costs are already included in the model. If no active decode request contributes a TPOT deadline, the allocator uses the remaining engine token capacity without an additional TPOT constraint. If $\Delta_k \leq 0$, it admits no prefill tokens in this step.

For a positive time budget, the allocator seeks the largest feasible prefill allocation:

$$p_k^* = \max_{p \in \mathbb{Z}_{\geq 0}} p, \\ \text{s.t. } p + d_k \leq B_{\text{max}}, \\ \hat{L}(p, d_k, K_k^{\text{dec}}, A_k^{\text{pre}}(p)) \leq \Delta_k. \quad (11)$$

Here, $A_k^{\text{pre}}(p)$ is computed from the candidate chunks in their scheduling order using Eq. 8. Once the decode workload and candidate order are fixed, only the admitted prefill token count varies, making the budget search one-dimensional. If even a decode-only step exceeds the selected time budget, the allocator sets the additional prefill allocation to zero and allows decode requests to continue making progress. The inequality provides a model-based scheduling target whose accuracy depends on the fitted predictor.

C. Token Allocation Workflow

Fig. 7 illustrates the interaction between the scheduler and allocator. At each iteration boundary, the scheduler collects the active decode workload, next-token deadlines, and current time (①). The allocator uses this state to establish the time budget and evaluate the step-time model as prefill candidates are considered (②). The scheduler processes ongoing requests and presents waiting prefill candidates in the order determined by the dispatcher, together with their chunk lengths and existing context lengths (③). For each candidate, the allocator combines these prefill statistics with the decode workload and checks both constraints in Eq. 11 to select a feasible chunk (④). The selected prefill tokens join the active decode tokens to form the mixed batch (⑤).

Running prefill chunks and newly admitted prefills share the same budget. As candidates are allocated, the allocator updates both the cumulative prefill token count and attention work. If the full chunk does not fit, it truncates the chunk to the largest feasible size; if no positive allocation remains, further prefill work waits for a subsequent step. The dashed segments in Fig. 7 represent this deferred work. Decode tokens remain scheduled subject to the engine’s resource limits.

In the example in Fig. 7, r_2 and r_3 remain active decode requests, while r_4 and r_5 are considered for prefill in that order. The allocator selects a feasible chunk from r_4 ’s prompt, leaving the remaining prefill work for later iterations. Thus, earlier admission does not require an entire prompt to fit in one step. After execution, updated token timestamps and context lengths determine the next allocation.

VII. IMPLEMENTATION OF TANGO

TANGO is implemented on top of vLLM 0.17.0 by extending its request admission path and scheduler. The dispatcher and allocator run in the CPU-side scheduling loop and produce per-request token allocations for each engine step. The resulting batch is passed to vLLM through its existing scheduler-executor interface, reusing continuous batching, chunked prefill, PagedAttention-based KV-cache management, tensor parallelism, sampling, and streaming output.

SLO Metadata and Runtime State: We extend request admission to carry two optional SLO fields, TTFT and TPOT, in milliseconds. A missing SLO is treated as an absent constraint, allowing best-effort requests to use the same serving path. Alongside these fields, the scheduler maintains each request’s arrival timestamp, prompt length, computed-token progress,

output-token count, and latest output timestamp in host memory. These fields are updated together with vLLM’s request state after each engine step. The first emitted token activates the TPOT objective, and subsequent token timestamps advance the next-token deadline.

Queue and Preemption Hooks: We add queue and preemption hooks to vLLM’s waiting-request admission path. The queue hook applies the phase-aware priority and tie-breaking rules in Section V. On KV-cache allocation failure, the preemption hook selects the running request with the latest deadline and requires the waiting request’s deadline to be strictly earlier. If permitted, vLLM reclaims the victim’s KV blocks, moves it from RUNNING to WAITING, and retries allocation. The victim keeps its SLOs, output progress, and the timestamps used to construct its deadline.

Per-Step Token Allocation: We fit the step-time model coefficients from offline profiling for the deployed model and hardware configuration. At each iteration boundary, the allocator summarizes the active decode workload and collects its next-token deadlines from scheduler-maintained state. It then establishes the time budget using the percentile policy in Section VI. As the scheduler processes ongoing prefill requests and admits waiting requests, their chunks share this budget and the engine’s maximum per-step token capacity. The allocator updates the cumulative prefill token count and attention work after each allocation, using the candidate’s chunk length and already-computed context length. A chunk that exceeds either limit is truncated to its largest feasible size; unallocated tokens remain for later steps. The resulting token allocations are forwarded to the executor through the standard scheduling output.

VIII. EVALUATION OF TANGO

In this section, we first evaluate the SLO attainment and goodput of TANGO across different models and multi-SLO workloads. We then investigate the effectiveness of its two scheduling components, validate the step-time model, examine sensitivity to sending rates and SLO settings, evaluate scalability with larger models and GPU deployments, and analyze scheduling overhead.

A. Evaluation Setup

Table II shows our experimental platform. The chatbot, mathematical-reasoning, and software-agent workloads are described in Section III-B. We use Llama-3.1-8B, Qwen3-14B, and Qwen3-32B (8B, 14B, and 32B for short) as base models. Based on the parameter size, we use 1, 2, and 4 NVIDIA RTX 6000 Ada GPUs to deploy them, respectively. Each RTX 6000 Ada has 48 GB of GPU memory. Moreover, we use four SLO classes (TTFT-prioritized, TPOT-prioritized, balanced, and relaxed) for each model.

We use the LLM serving systems vLLM, SOLA, and JITServe as baselines. vLLM employs FCFS admission and a fixed maximum per-step token budget, without differentiating request-specific TTFT and TPOT objectives. SOLA adjusts both request execution order and iteration workload according

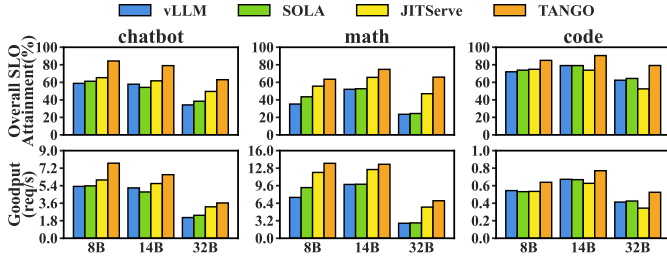


Fig. 8. The overall SLO attainment and goodput of TANGO, vLLM, SOLA, and JITServe in various scenarios.

to request and system states to improve SLO attainment. JITServe estimates uncertain request characteristics and allocates serving bandwidth to individual requests, while grouping compatible requests to improve batching efficiency. Since SOLA’s source code is not publicly available and JITServe uses a different vLLM version, we reimplement SOLA’s scheduling mechanisms and port those of JITServe to vLLM 0.17.0. This allows all systems to be compared on the same vLLM runtime.

Following prior works, we use overall SLO attainment and goodput as performance metrics. Overall SLO attainment is the percentage of all requests that satisfy both their TTFT and TPOT SLOs. Goodput measures the number of such requests completed per second.

B. SLO Attainment and Goodput

We first evaluate TANGO on overall SLO attainment and goodput. Each workload is evaluated on the 8B, 14B, and 32B models. For each model–workload combination, we replay the trace at $1.2\lambda_{90}$, where λ_{90} is the sending rate at which vLLM achieves 90% SLO attainment for that combination. All systems use the same request sequence, arrival schedule, and SLO targets. Fig. 8 shows the results for all nine model–workload combinations.

As observed, TANGO achieves the highest overall SLO attainment and goodput in all test cases. Across the nine cases, the average increases in SLO attainment are 23.37%, 21.52%, and 15.48% compared with vLLM, SOLA, and JITServe, respectively. The corresponding average goodput ratios are $1.52\times$, $1.47\times$, and $1.22\times$. The most pronounced combined improvement occurs in the Qwen3-32B mathematical-reasoning case, where TANGO increases overall SLO attainment by 42.40%, 41.50%, and 18.90% over vLLM, SOLA, and JITServe, respectively, while the corresponding goodput ratios are $2.50\times$, $2.44\times$, and $1.20\times$.

These improvements are consistent with TANGO’s joint control of request urgency and step duration. The dispatcher prioritizes earlier deadlines for admission and preempts a later-deadline running request when KV-cache allocation fails. The allocator uses predicted step time and active latency slack to bound the prefill work sharing a step with decode requests. This coordination addresses the TTFT violations from queueing and TPOT violations from mixed prefill/decode execution identified in Section III-D.



Fig. 9. The goodput and overall SLO attainment of TANGO’s variants normalized to TANGO, represented by bars and curves, respectively.

C. Effectiveness of Temporal Scheduling

In this subsection, we show the performance of TANGO-WOT, a variant of TANGO that disables temporal scheduling. It restores vLLM’s default admission and preemption policies, including FCFS request admission, while retaining the step-time-model-guided token allocator. All other experimental conditions, including the nine model–workload combinations and sending rates, are kept the same as in Section VIII-B.

The green bars and curves of Fig. 9 show the goodput and overall SLO attainment of TANGO-WOT normalized to TANGO. As observed, both metrics decrease in all nine cases, with average relative reductions of 20.31% and 20.39%, respectively.

Without phase-aware prioritization, FCFS can leave requests with tight TTFT targets waiting behind earlier requests with more latency slack. The default preemption policy also does not compare active deadlines when deciding which running request should yield KV-cache capacity. Although the retained allocator regulates the execution time of mixed batches, it cannot by itself give earlier admission to urgent waiting requests. Consequently, controlling step duration alone is insufficient to recover the SLO attainment and goodput of the complete system.

D. Effectiveness of Spatial Scheduling

In this subsection, we show the performance of TANGO-WOS, a variant of TANGO that disables spatial scheduling. It replaces model-guided token allocation with a fixed maximum token budget per step, while retaining the deadline-aware request dispatcher. Thus, request admission and preemption still use phase-aware deadlines, and all other experimental conditions remain unchanged.

The blue bars and curves of Fig. 9 show the goodput and overall SLO attainment of TANGO-WOS normalized to TANGO. Both metrics remain below those of TANGO in every case, with average relative reductions of 13.19% in goodput and 12.66% in SLO attainment.

A fixed token budget does not provide a reliable bound on the duration of a mixed prefill/decode step. As batch composition and context lengths change, the same token count can incur different computation and KV-cache access costs. A long prefill chunk can therefore delay the next tokens of decode requests even when request admission follows deadline order. The model-guided allocator controls this interference by adjusting the prefill allocation to the active decode

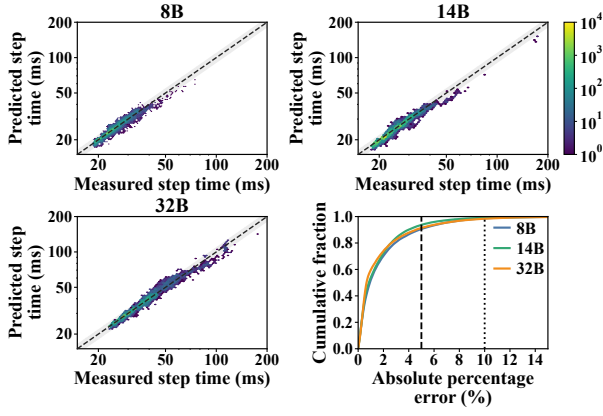


Fig. 10. Step-time prediction accuracy for the three models. Diagonal dashed lines denote perfect predictions; color indicates step counts per hexagon (log scale). The bottom-right panel shows absolute percentage error CDFs.

workload and remaining latency slack. These results show that deadline-aware request ordering and execution-time-aware token allocation make complementary contributions to serving performance.

E. Accuracy of the Step-Time Model

In this subsection, we evaluate the prediction accuracy of the step-time model in Eq. 7. For each deployment configuration, we compare the predicted duration $\hat{L}$ with the measured duration L using its fitted coefficients. The evaluated steps cover decode-only and mixed execution across the three workloads at low and high loads. We evaluate 32,542, 30,988, and 21,952 steps for the 8B, 14B, and 32B models, respectively. We measure prediction error as the absolute percentage error $100|\hat{L} - L|/L$.

Fig. 10 shows predicted versus measured step times and absolute percentage error CDFs (bottom right). As observed, predictions closely match measurements across all three models, with errors concentrated at small values. The mean absolute percentage errors (MAPEs) are 1.83%, 1.62%, and 1.70% for the 8B, 14B, and 32B models, respectively. The fractions of steps with errors below 5% are 90.59%, 93.70%, and 91.38%, respectively. At 10%, the corresponding fractions reach 98.68%, 99.05%, and 98.36%.

F. Sensitivity to the Sending Rate

In this subsection, we investigate the performance of TANGO under different sending rates. The overall comparison in Section VIII-B uses $1.2\lambda_{90}$ to evaluate serving performance under pressure. Here, we vary the load multiplier α from 1.0 to 1.4 in increments of 0.05 and replay the trace at $\alpha\lambda_{90}$. We use Qwen3-14B with the chatbot workload for this experiment. All systems use the same request sequence, arrival schedule, and SLO targets at each sending rate.

Fig. 11 shows the goodput and overall SLO attainment of TANGO and the baselines. At λ_{90} , without additional stress, all four systems achieve high SLO attainment and similar goodput, with TANGO still leading in both metrics. As the sending

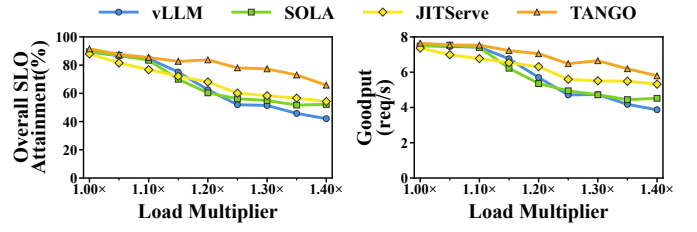


Fig. 11. The goodput and overall SLO attainment of TANGO, vLLM, SOLA, and JITServe under different sending rates for Qwen3-14B on the chatbot workload. The load multiplier is relative to λ_{90} .

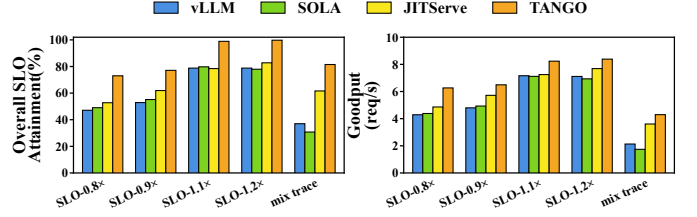


Fig. 12. The overall SLO attainment and goodput of TANGO, vLLM, SOLA, and JITServe under different SLO settings.

rate increases, both metrics generally decline, but TANGO maintains higher SLO attainment and goodput, showing a more pronounced advantage under pressure. These results are consistent with TANGO’s coordinated control of request ordering and step duration. As the sending rate increases, more requests compete for execution, making queueing delay and prefill/decode interference harder to control. The dispatcher prioritizes urgent requests for admission, while the allocator adjusts the prefill work in each step according to the predicted duration and active latency slack. This coordination helps TANGO preserve higher SLO attainment and goodput under increasing pressure, although both metrics also decline at higher loads. The results show that TANGO’s performance benefits persist across the evaluated sending rates.

G. Sensitivity to SLO Settings

In this subsection, we evaluate TANGO under different SLO settings. For Qwen3-14B on the chatbot workload, we scale both TTFT and TPOT targets by $\eta \in \{0.8, 0.9, 1.1, 1.2\}$. We also combine requests from different trace types into a mixed workload. All requests of the same trace type share one fixed TTFT/TPOT SLO configuration, while different types can use different configurations. All systems use identical requests and arrivals within each setting.

Fig. 12 shows that TANGO achieves the highest overall SLO attainment and goodput in all five settings. Across the four SLO scaling factors, the average increases in SLO attainment are 22.80%, 21.72%, and 18.22% compared with vLLM, SOLA, and JITServe, respectively. The corresponding average goodput ratios are $1.29\times$, $1.28\times$, and $1.16\times$.

For the mixed trace, TANGO increases SLO attainment by 44.50%, 50.75%, and 19.88% over vLLM, SOLA, and JITServe, respectively, with corresponding goodput ratios of

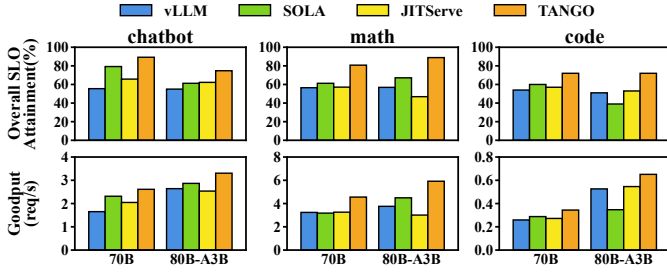


Fig. 13. The overall SLO attainment and goodput of TANGO, vLLM, SOLA, and JITServe with the 70B and 80B-A3B models on the three workloads.

2.02 $\times$, 2.47 $\times$, and 1.19 $\times$. These results show that deadline-aware request ordering and step-time-guided token allocation remain effective under both SLO scaling and mixed workloads.

H. Scalability with Larger Models and More GPUs

In this subsection, we investigate the scalability of TANGO with larger models and increased GPU resources. We use Llama-3.1-70B and Qwen3-Next-80B-A3B-Instruct (70B and 80B-A3B for short), with the latter using a mixture-of-experts (MoE) architecture. Both models are deployed on eight NVIDIA A800 GPUs with a tensor-parallel degree of 8. We evaluate them on the chatbot, mathematical-reasoning, and software-agent workloads, using the same sending-rate and SLO settings as in Section VIII-B.

Fig. 13 shows the overall SLO attainment and goodput of TANGO and the baselines. As observed, TANGO achieves the highest values for both metrics in all six model-workload combinations. The average increases in SLO attainment are 24.79%, 18.29%, and 22.60% compared with vLLM, SOLA, and JITServe, respectively. The corresponding average goodput ratios are 1.40 $\times$, 1.35 $\times$, and 1.40 $\times$. The above results demonstrate that TANGO remains effective when scaling to larger dense and MoE models, as well as deployments with more GPUs and greater aggregate GPU memory.

I. Overhead of TANGO

Time Overhead: It mainly comes from deadline maintenance, request ordering, and preemption decisions in the dispatcher, and decode-deadline selection, step-time prediction, and prefill-budget search in the allocator. Both components run on the CPU before model execution and reuse vLLM’s existing executor. The closed-form step-time model requires only a few arithmetic operations for each prediction. For a scheduling step with thousands of candidate requests, the estimated CPU time for deadline maintenance and decode-deadline selection is less than 0.5 ms in total, while step-time prediction and budget search take less than 0.1 ms. Since model execution typically takes tens to hundreds of milliseconds per step, this estimated overhead is usually below 1% of the step latency.

Memory Overhead: TANGO stores additional scheduling state as CPU-side metadata, including request SLOs, arrival and latest-token timestamps, generated-token counts, effective deadlines, and progress information. A compact representation

is estimated to require approximately 96 bytes of persistent metadata per request, corresponding to about 240 KB for 2,500 resident requests. Moreover, the allocator stores the fitted step-time coefficients and a few configuration values, with a total footprint below 128 bytes. Its temporary decode-deadline list requires about 16 bytes per active decode request. The per-request state scales linearly with the request count and is negligible compared with the memory occupied by model weights and KV caches.

IX. CONCLUSION

In this paper, we propose TANGO to improve multi-SLO LLM serving through joint temporal and spatial scheduling. Its dispatcher uses phase-aware deadlines to guide request admission and preemption according to heterogeneous TTFT and TPOT requirements. Its allocator uses predicted step time and remaining decode slack to allocate prefill tokens, limiting mixed prefill/decode interference. The evaluation results show that TANGO improves overall SLO attainment by 58.72% and achieves 1.52 $\times$ the goodput on average compared with state-of-the-art systems.